

Haoming Li

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EDUCATION

The Hong Kong University of Science and Technology

Sep 2022 – May 2026 (Expected)

B.Eng. in Computer Science | **GPA: 3.343 / 4.300**

HKSAR, China

- **Core Coursework:** Design and Analysis of Algorithms (A), Operating Systems (A+), Discrete Mathematical Tools for Computer Science (A+), Object-Oriented Programming and Data Structures (A-), Introduction to Multivariable Calculus (A+), Matrix Algebra and Application (A), Explore Artificial Intelligence, Mobile Application Development, Internet Computing, Multimedia Computing, Advanced Topics in Multimodal Machine Learning (graduate-level)
- **Honors and Awards:** University's Scholarship Scheme for Continuing Undergraduate Students (2024-2025, TOP 10% CGA), Dean's List of the School of Engineering (2023-2024 TGA ≥ 3.7), University's Scholarship Scheme for Continuing Undergraduate Students (2023-2024, TOP 10% CGA), Dean's List of the School of Engineering (2022-2023 TGA ≥ 3.7)

National University of Singapore

Aug 2026 – TBD

Master of Computing (Artificial Intelligence)

INTERNSHIP AND RESEARCH EXPERIENCE

Lively Impact Technology Limited

Jul 2025 – Dec 2025

Software Development Intern

HKSTP (Hong Kong Science Park)

- Contributed to the development of a web-based AI storytelling platform, implementing an end-to-end workflow for text-to-image, image-to-image, image-to-video generation, video merging and related competition system, assessment system, users management system.
- Designed and built an interactive web-based canvas interface using Vue.js and Vuetify, featuring draggable, linkable, and dynamically updating nodes for story flow visualization, enhancing usability and creative flexibility.
- Integrated AI model APIs with multithreaded task scheduling to improve image and video generation efficiency.
- Developed advanced features including multilingual voice synthesis (English, Chinese, Cantonese), background music, subtitle generation, and an AI-driven prompt scoring and optimization system.

Visual-Audio Editing (Audio-Visual Object Removal)

Sep 2024 – Dec 2024

- Constructed training datasets for audio-visual object removal by generating mask videos using SAM2 segmentation, extracting corresponding audio, and mixing with background sounds to ensure natural audio-visual consistency.
- Reviewed and applied insights from key literature to design the dataset preparation and audio separation methodology.
- Modified SAM2 code to support MP4 input/output, mouse-based selection for target objects, client-server JSON workflow, and efficient batch processing, overcoming limitations of the original PNG-only implementation.
- Debugged and optimized technical issues using Python; set up environments with WSL, integrated deep learning models, and processed multimodal data with FFmpeg workflows.

PROJECT EXPERIENCE

QUST – The Academia Questing App | COMP4521 Mobile Application Development

Mar 2025 – Jun 2025

- Developed a gamified task management mobile app using Kotlin and Jetpack Compose, integrating RPG-style quests to enhance user engagement, habit building, and academic support.
- Implemented a real-time client-server architecture with Firebase Firestore, Cloud Functions, and Authentication, enabling dynamic quest generation, user-specific rewards, and real-time progress updates.
- Used Gemini AI to score quests and assign rewards, and provided personalized feedback.
- Designed interactive front-end interfaces, including dashboard, quest log, social hub, and profile customization, supporting real-time notifications, collaborative quest sharing, and gamified progress visualization.

Maze Fight Game | COMP4021 Internet Computing

Oct 2024 – Dec 2024

- Developed a real-time two-player maze game with registration/login system, dynamic countdown, and smooth WASD-based movement for interactive gameplay.
- Implemented a complex item system to enable strategic combat, health management, and instant teleportation.
- Engineered core game logic, including timed matches, HP-based win conditions, tie resolution, and cheat detection.
- Built responsive front-end interfaces and comprehensive game-over pages featuring dynamic leaderboards, detailed win/loss statistics, real-time score updates, and fully functional replay and quit options, enhancing user experience.

Video Stabilization | *COMP4431 Multimedia Computing*

Apr 2024 – May 2024

- Developed a video stabilization system to transform shaky raw footage into smooth visual output.
- Implemented a brute-force search algorithm to determine optimal block movements between consecutive frames, minimizing mean absolute error (MAE) for accurate alignment.
- Designed and applied motion vector smoothing filters across multiple frames, recombining blocks with inferred positions and using original frames to fill missing pixels for seamless stabilization.
- Built an interactive front-end with features including video upload, customizable block size, selectable smoothing filters, motion vector visualization toggle, pixel fill options, and stabilized video generation, enhancing usability and technical demonstration of stabilization algorithms.

A Survey about Vision-Enabled Large Language Models

Apr 2025 – May 2025

6411C Advanced Topics in Multimodal Machine Learning

- Conducted a literature survey on state-of-the-art vision-language models, including CLIP, LLaVA, and Kosmos-2, analyzing model architectures, feature extraction methods, training strategies, datasets, and limitations.
- Investigated key challenges such as multimodal hallucination, high-resolution image processing, long-context reasoning, and cross-modal alignment, proposing insights for future model improvements.
- Applied critical analysis to compare model performance across benchmarks and instruction-following capabilities, highlighting trends and opportunities for general-purpose multimodal AI systems.
- Produced a detailed survey report and a presentation summarizing advancements in multimodal integration, instruction tuning, and benchmark evaluation for large language models with visual input.

OTHERS

Technical Skills: C++, C, Python, Java

Languages: Mandarin (native), English (TOEFL 102)